FACULTY OF ARTS

KU LEUVEN

Computational Linguistics

Lecture 5: Machine translation

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Tuesday 3 November 2026

Machine translation

- Machine translation is a difficult task
- Ultimate test for language understanding
- In order to translate correctly, one needs to
  - understand the **source** language
  - understand the **target** language
  - have knowledge about the **topic** of the text to be translated
  - have knowledge about the **culture**, values, traditions, expectations of language users in both languages







A long history

- Warren Weaver (1949):
  "I have a text in front of me which is written in Russian, but I am going to pretend that it is really written in English, and that it has been coded in some strange symbols. All I need to do is strip off the code in order to retrieve the information contained in the text."
- First system demonstration by IBM in 1954 (word by word translation)
- But machine translation turned out to be much more difficult than expected



Political/commercial interest

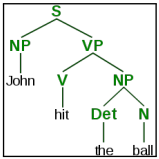
- The EU spends more than a billion euros on translation costs each year; (semi-)automation would save a lot of money
- The US invested a lot in machine translation for intelligence acquisition
- Google Translate is used by more than a billion people every month and supports close to 250 languages
- Today, translation is built into browsers, phones, and chat assistants





Different approaches

- **Symbolic** machine translation
  - Based on linguistics and logic
- **Statistical** machine translation
  - Based on Bayes' theorem
- **Neural** machine translation
  - Based on neural networks

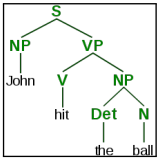






Different approaches

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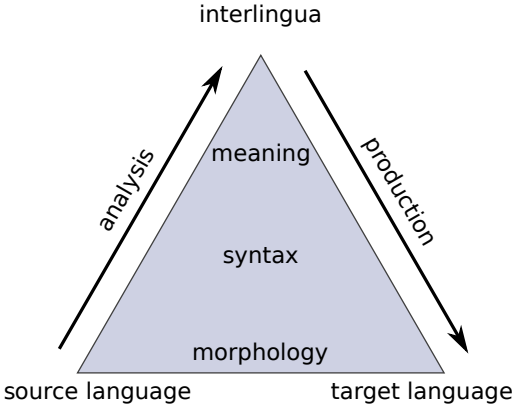






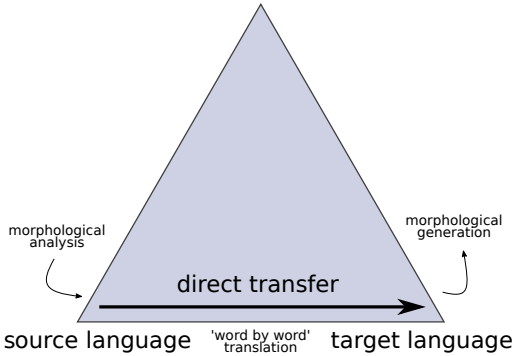
Symbolic machine translation

Vauquois triangle



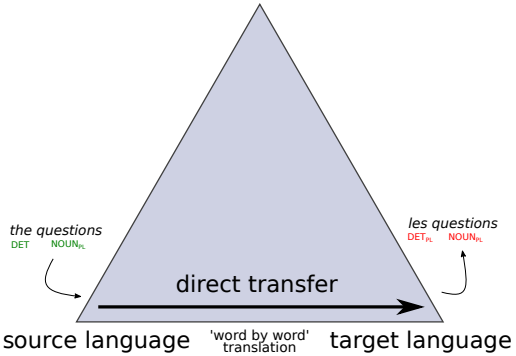
Symbolic machine translation

Direct transfer



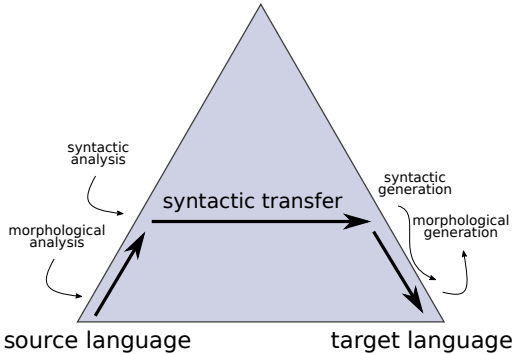
Symbolic machine translation

Direct transfer



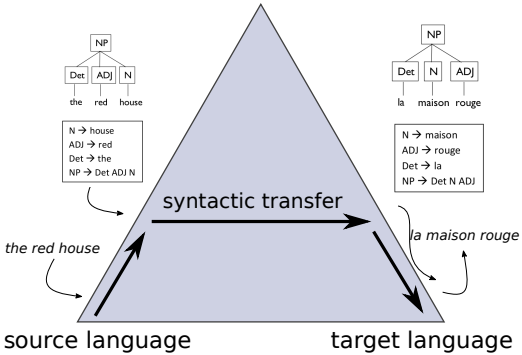
Symbolic machine translation

Syntactic transfer



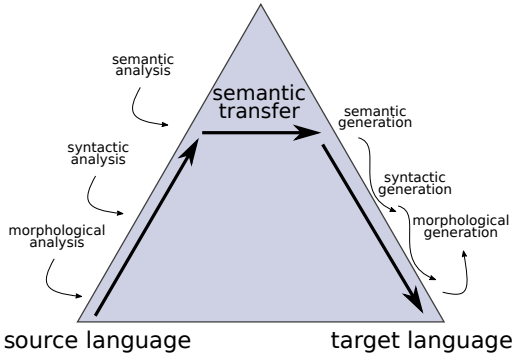
Symbolic machine translation

Syntactic transfer



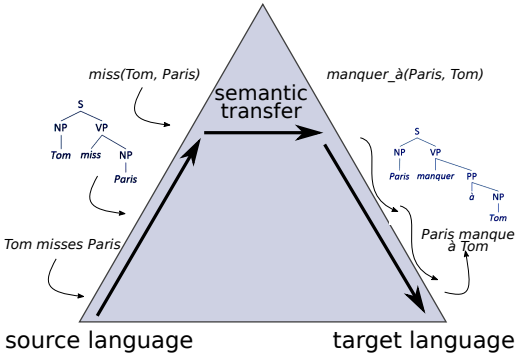
Symbolic machine translation

Semantic transfer



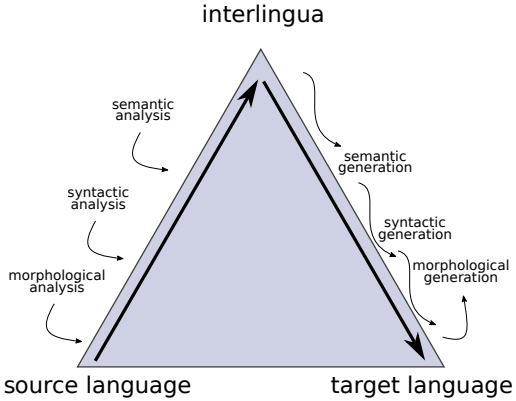
Symbolic machine translation

Semantic transfer



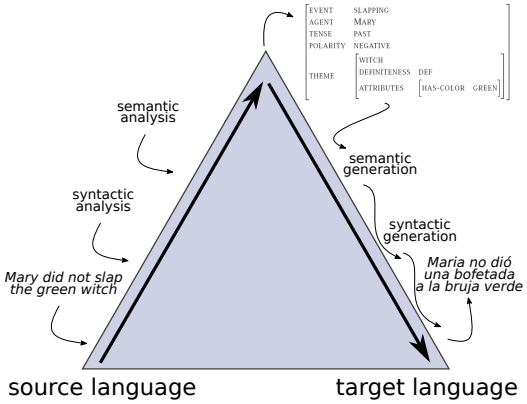
Symbolic machine translation

Interlingua



Symbolic machine translation

Interlingua



Symbolic machine translation

Inconveniences

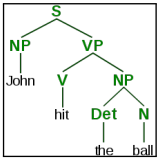
- Symbolic machine translation works well for applications within a restricted domain
- Difficult to generalize to open domain (ambiguities, exceptions, ...)
- Creation of rules is done by hand
- Tedious and costly development process



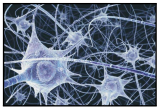


Different approaches

- Symbolic machine translation
  - Based on linguistics and logic
- **Statistical machine translation**
  - **Based on Bayes' theorem**
- Neural machine translation
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Statistical machine translation

Probabilities

- Find the most likely sentence in English given a sentence in French

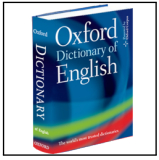
$$\begin{aligned}\hat{e} &= \underset{e \in english}{\operatorname{argmax}}\, p(e \mid f) \\ &= \underset{e \in english}{\operatorname{argmax}}\, \frac{p(f \mid e)p(e)}{p(f)} \\ &= \underset{e \in english}{\operatorname{argmax}}\, p(f \mid e)p(e)\end{aligned}$$



Statistical machine translation

Why Bayes ?

- Why don't we model $p(e \mid f)$?
- By using $p(f \mid e)P(e)$, we can divide the work
  - $p(e)$ deals with correct usage of English
  - $p(f \mid e)$ makes sure that French corresponds to English
  - We can train both models independently





Statistical machine translation

Analogy of a crime scene

- f is a crime scene, and we are looking for a suspect e
  - $p(e \mid f)$: 'Who did it?'
  - $P(e)$: Who had a motive?
  - $P(f \mid e)$: Could they have done it (crime scene investigation, alibi) ?
- Someone can have a good motive but not the means – both are necessary



Statistical machine translation

Example

Nous regardons la télévision

	correct English usage? $P(E)$	good correspondence French/English? $P(F \mid E)$
We look the television	×	✓
It bike nothing bottle	×	×
In television we watch	×	✓
That seems fairly reasonable	✓	×
We are watching television	✓	✓
Are watching television we	×	✓
He felt relieved	✓	×

Statistical machine translation

Example

Nous regardons la télévision

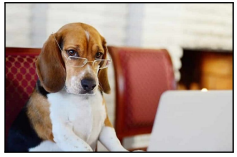
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That seems fairly reasonable	✓	×
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Are watching television we	×	✓
He felt relieved	✓	×

Statistical machine translation

$p(e)$ – correct usage

How to compute p(the,dog,was,running,across,the,green,field) ?

- We compute the joint probability by decomposing into conditional probabilities
- $p(\text{the}) \times p(\text{dog}|\text{the})$
  $\times\ p(\text{was}|\text{the dog})$
  $\times\ p(\text{running}|\text{dog was})$
  $\times\ p(\text{across}|\text{was running})$
  $\times\ p(\text{the}|\text{running across})$
  $\times\ p(\text{green}|\text{across the})$
  $\times\ p(\text{field}|\text{the green})$
- ... using only a limited context



Statistical machine translation

Rosetta stone

- During centuries, the Egyptian language was a mystery
- In 1799, a stone was discovered with an Egyptian text and its translation in Greek
- Enabled humans to learn to translate Egyptian



Statistical machine translation

$p(f \mid e)$ – good correspondence

- Our algorithms will use the same idea
- We use a large collection of parallel texts in two languages: **parallel corpus**
- Today's Rosetta stone :
  - Proceedings of European Parliament
  - Movie subtitles
  - Multilingual web sites
  - ...
- We know which **sentences** correspond, but not which **words**







Statistical machine translation

Intuition

What is the word for *heaven* in Vietnamese ?

English In the beginning God created the **heavens** and the earth.
Vietnamese Ban dâu Dúc Chúa Trời dung nên trời dât.

English God called the expanse **heaven**.
Vietnamese Dúc Chúa Trời dat tên khoang không la trời.

English ...you are this day like the stars of **heaven** in number.
Vietnamese ...các nguoi dông nhu sao trên trời.

Statistical machine translation

Intuition

What is the word for *heaven* in Vietnamese ?

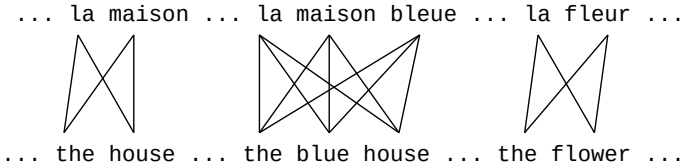
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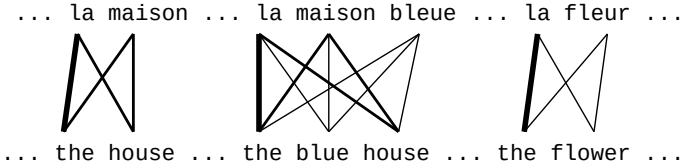
Statistical machine translation

Alignment



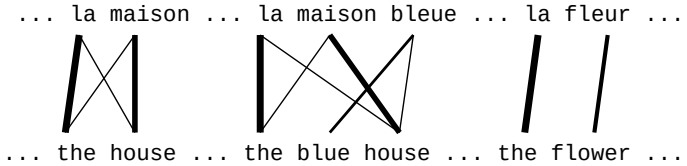
Statistical machine translation

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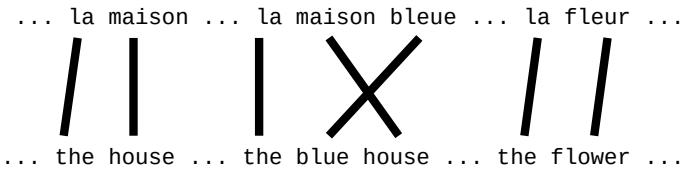
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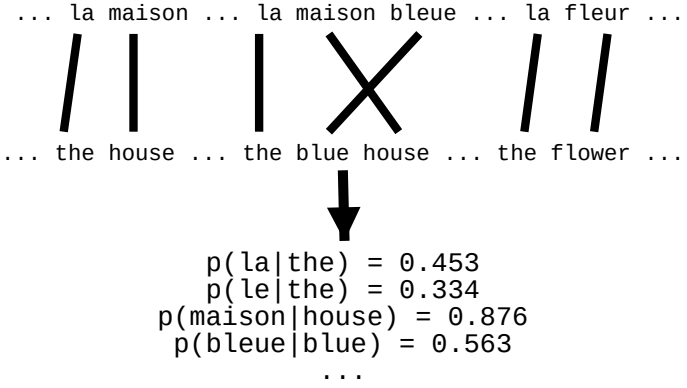
Statistical machine translation

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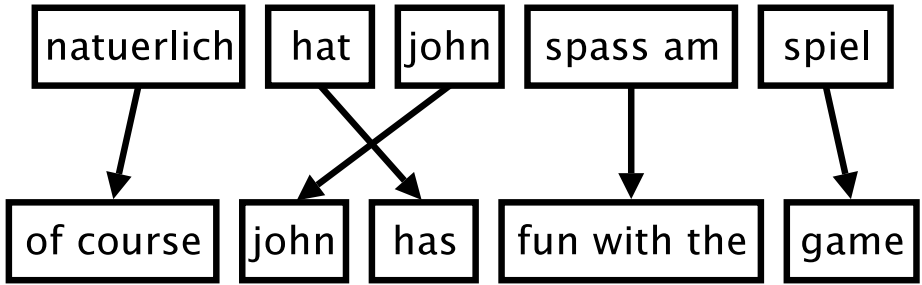
Statistical machine translation

Alignment



Statistical machine translation

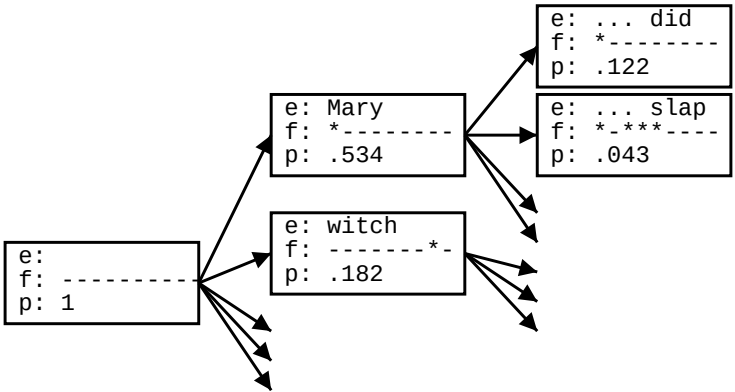
Alignment



- Alignment between units consisting of multiple words

Statistical machine translation

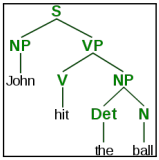
Decoding



- Construction of the sentence, from left to right
- Reduction by pruning weak hypotheses

Different approaches

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- **Neural machine translation**
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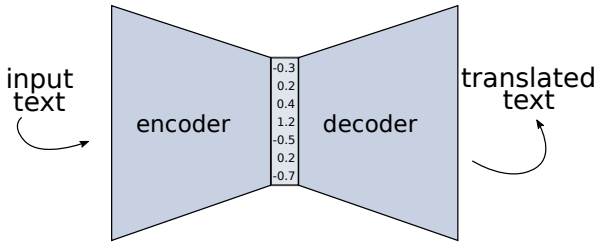




Neural machine translation

Concept

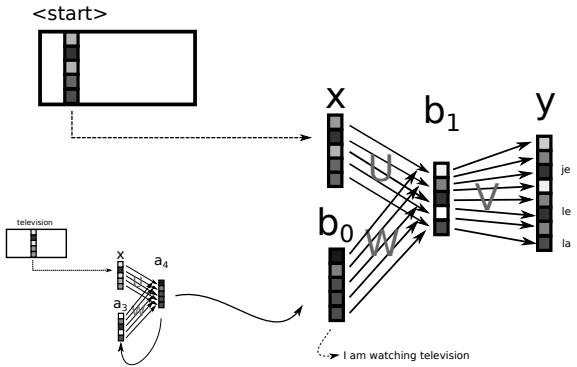
- Model translation process as one big mathematical function: a neural network



- **Encoder-decoder** architecture with attention: an RNN in early systems, a transformer today
- Trained end-to-end on **parallel corpora**: the same data as statistical MT, but a single model instead of a pipeline of components

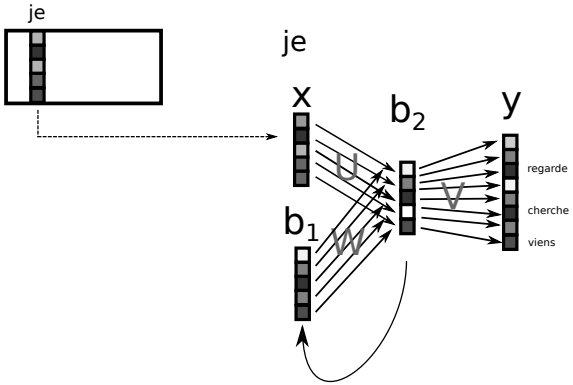
Recurrent neural network

Encoder-decoder



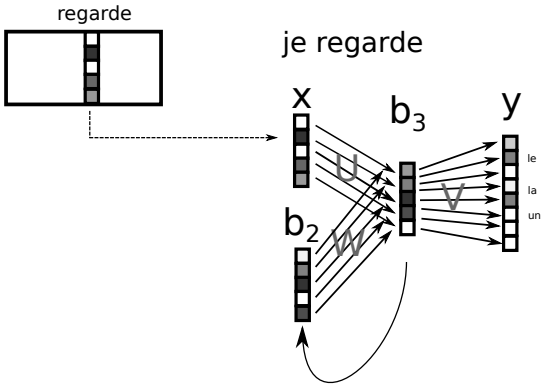
Recurrent neural network

Encoder-decoder



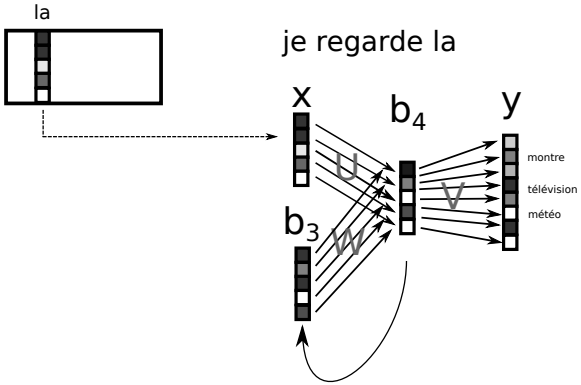
Recurrent neural network

Encoder-decoder



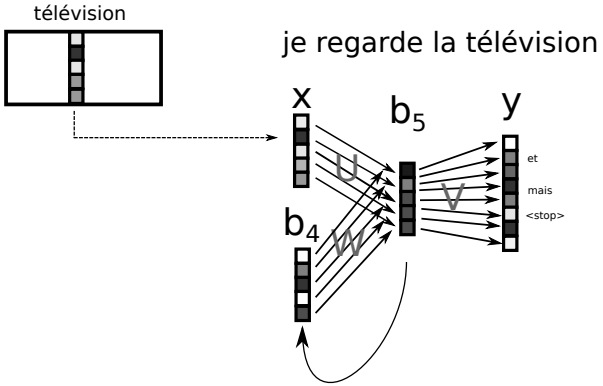
Recurrent neural network

Encoder-decoder

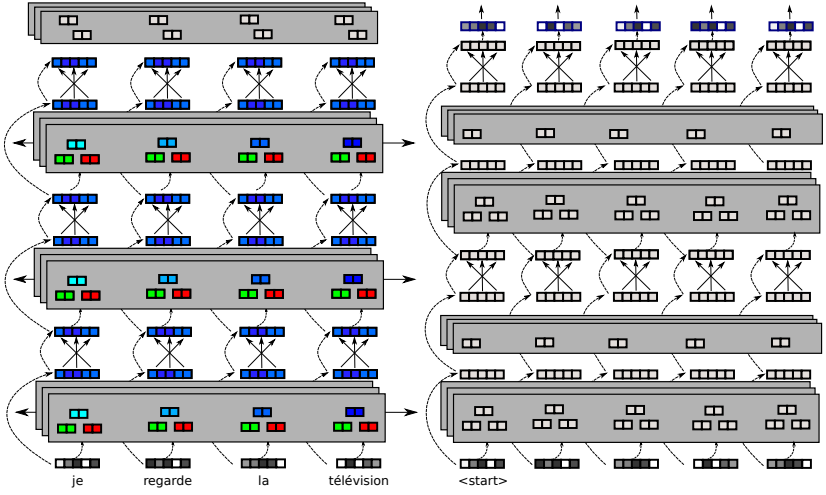


Recurrent neural network

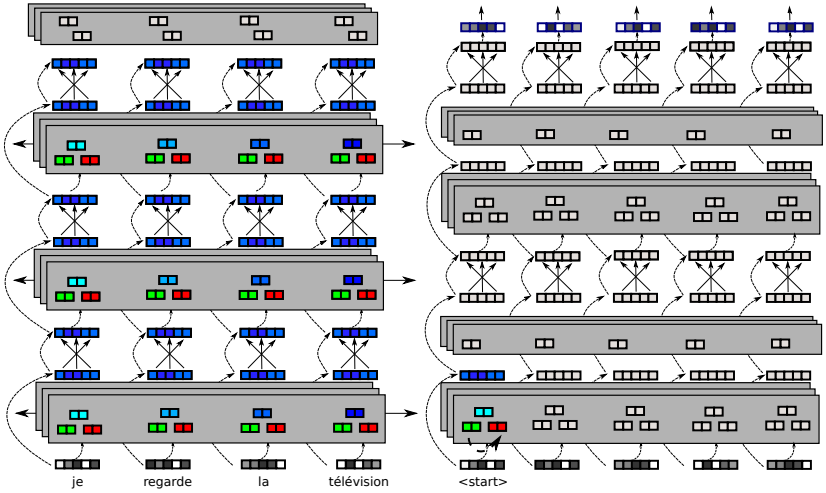
Encoder-decoder



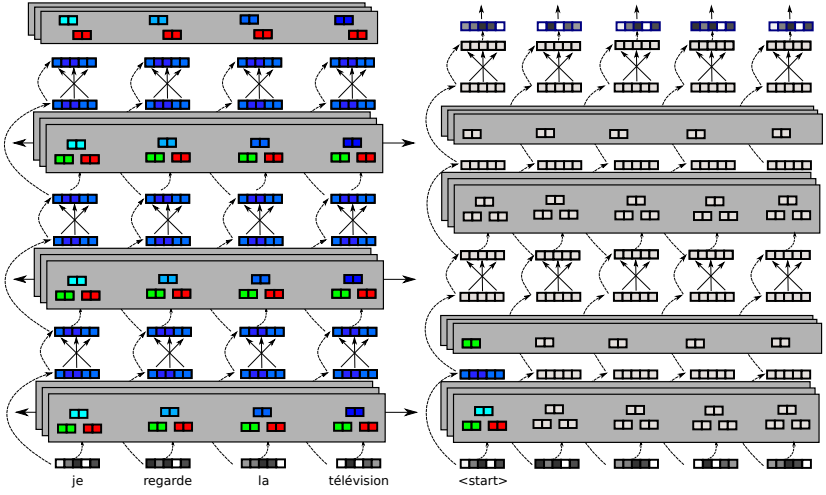
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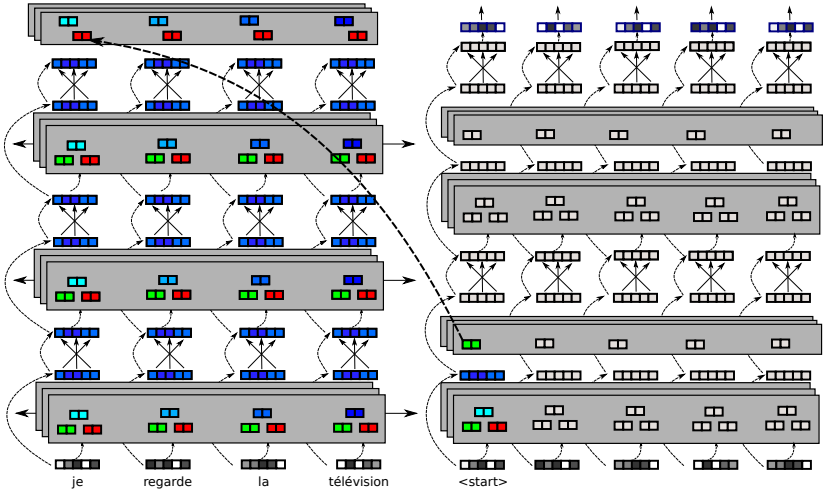
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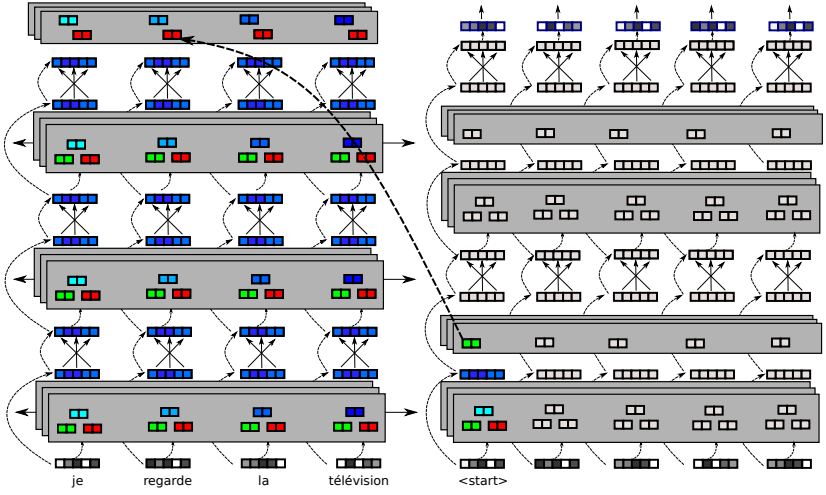
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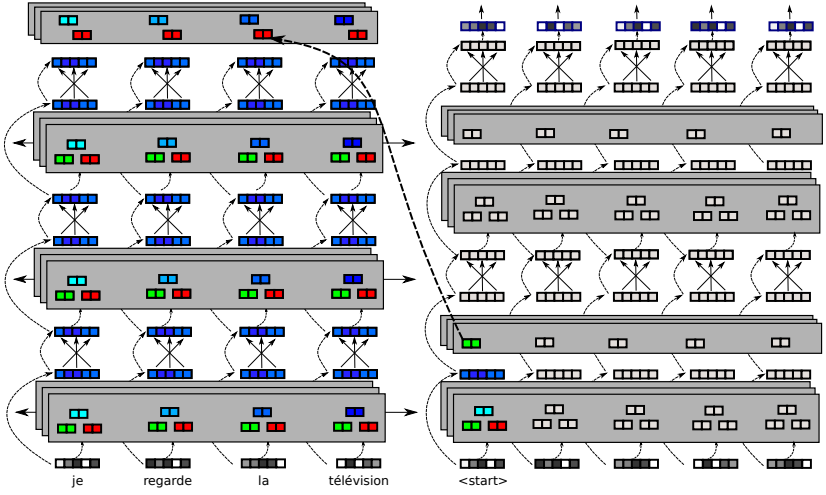
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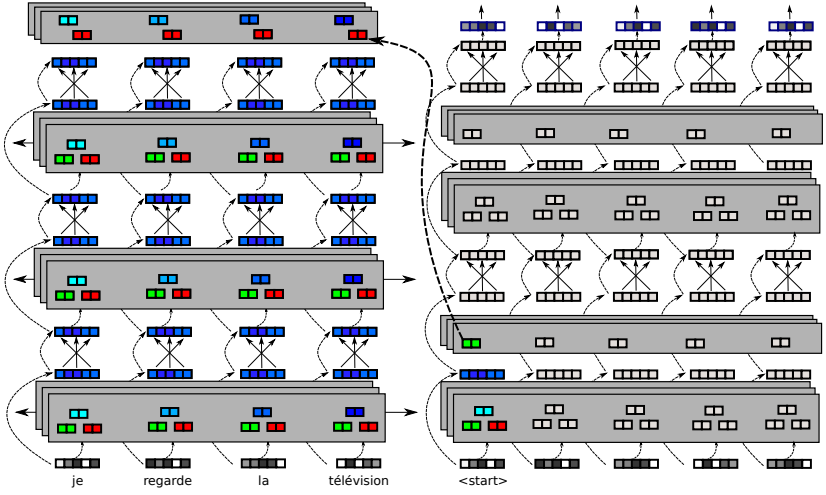
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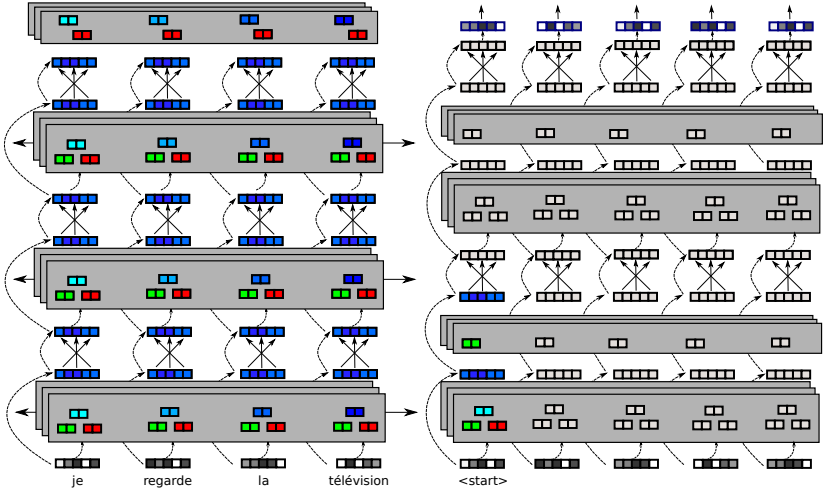
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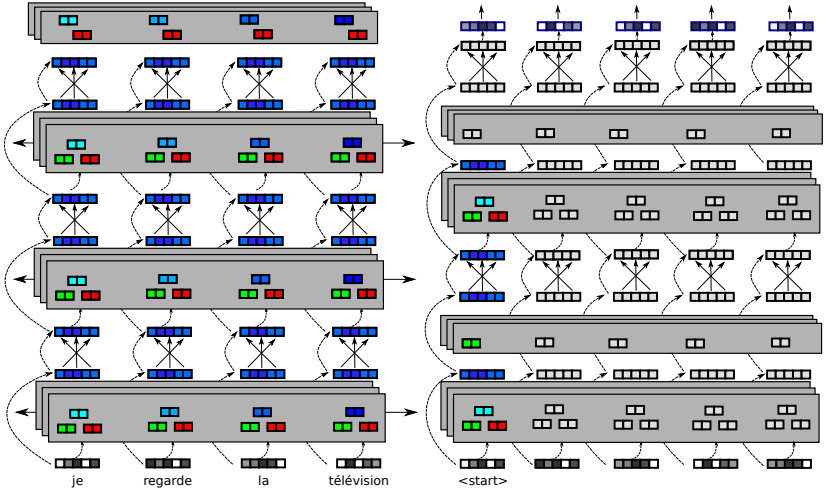
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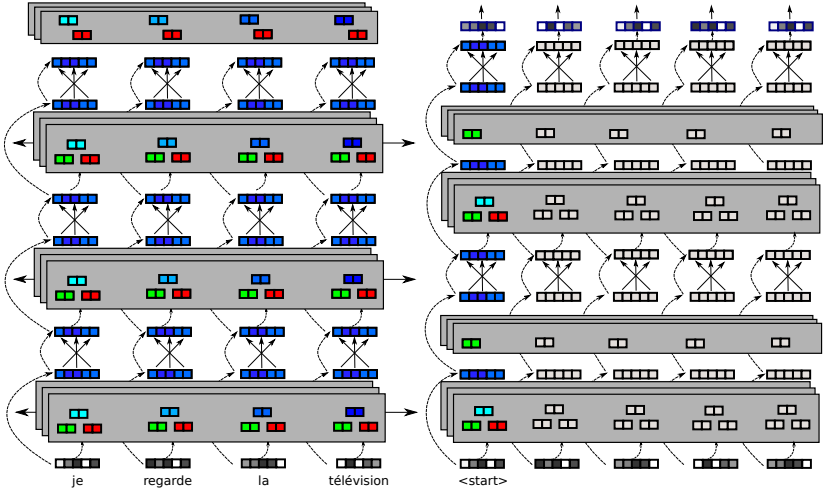
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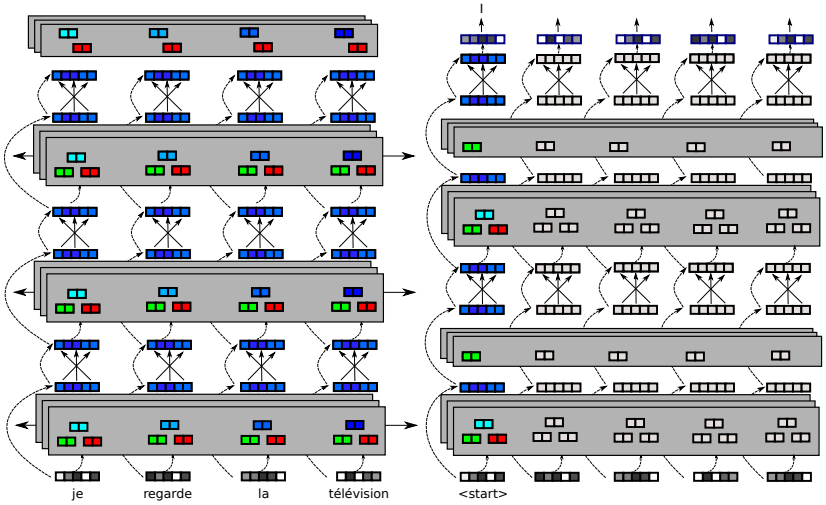
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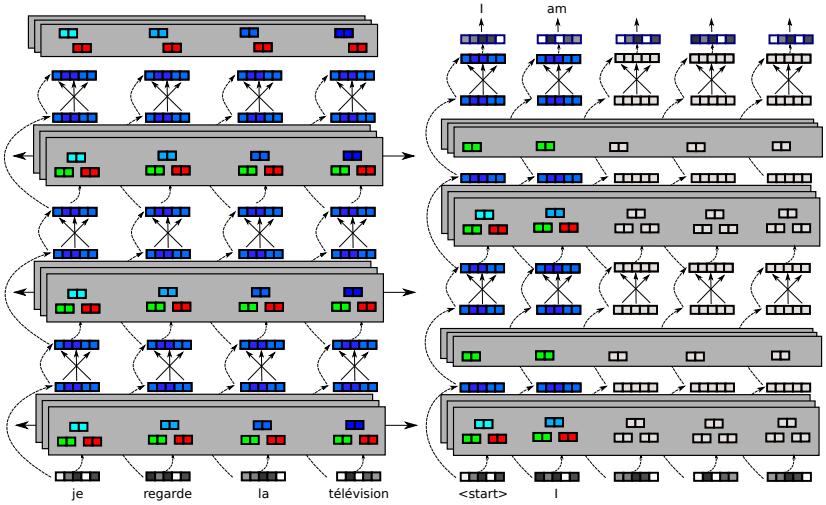
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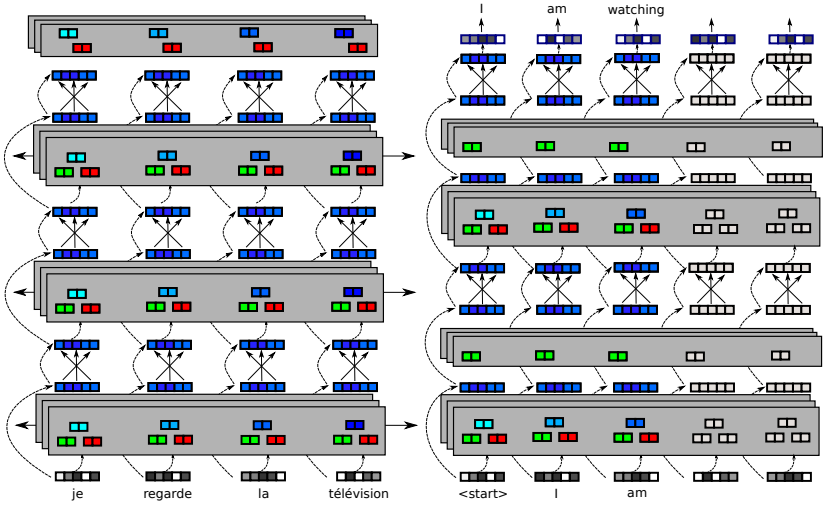
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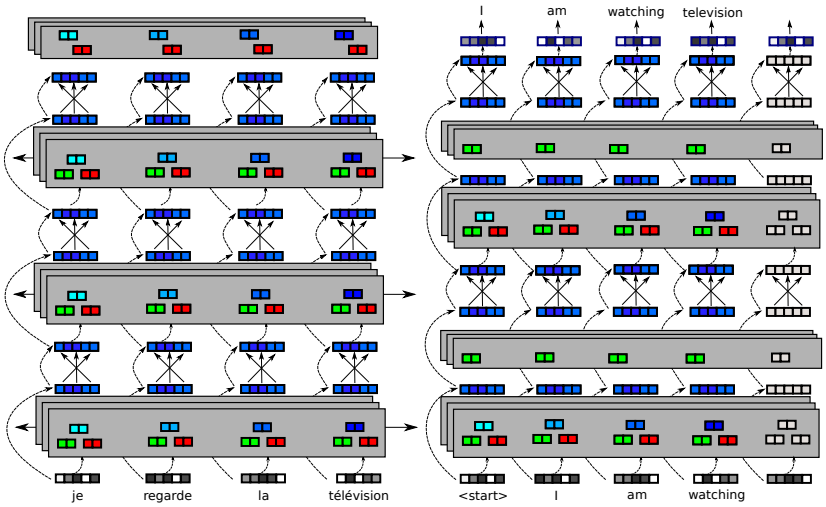
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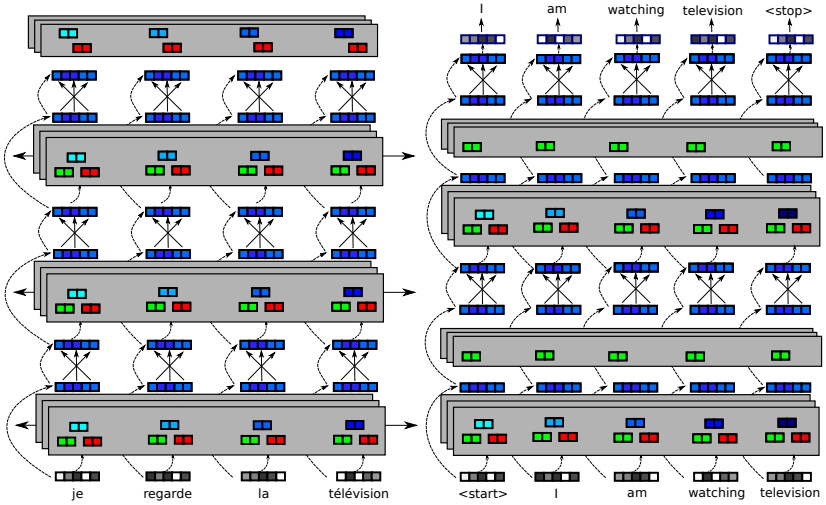
Encoder-decoder



Encoder-decoder



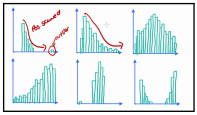
Encoder-decoder



Neural machine translation

Recent history

- Statistical machine translation represented the state of the art until 2015
- 2014: first sequence-to-sequence neural translation models (RNN-based)
- 2016: Google Translate switches to neural; NMT widely adopted as the privileged method
- 2017: the **transformer**, originally developed for machine translation, becomes the standard architecture
- 2020s: massively multilingual models; translation as one of the many tasks of large language models





Neural machine translation

Translation quality

- Two dimensions of translation quality:
  - **adequacy**: does the translation preserve the meaning of the source?
  - **fluency**: is the output natural, grammatical target language?
- Statistical MT: often adequate, less fluent (concatenation of translated phrases)
- Neural MT: considerably more **fluent**; better handling of word order and morphology
- One end-to-end model replaces a pipeline of separately trained components (alignment, phrase table, language model, reordering)
- Fluent output can conceal adequacy errors (omissions, additions, invented content)

Multilingual machine translation

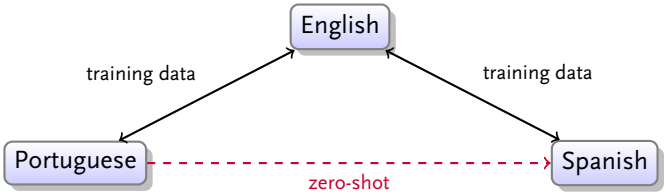
- Classical setup: one model per language pair and direction
- N languages $\rightarrow N(N-1)$ translation directions
  - EU: 24 official languages $=$ 552 directions
- For most pairs, hardly any parallel data exists (e.g. Finnish $\leftrightarrow$ Swahili)
- Alternative: a **single model** trained on many languages at once

Multilingual NMT

- One shared encoder-decoder for many languages
- Target language indicated by a special tag
  - `<2nl>` How are you? $\rightarrow$ *Hoe gaat het?*
  - `<2fr>` How are you? $\rightarrow$ *Comment ça va?*
- All languages share the same parameters and subword vocabulary
- **Transfer** between related languages, in particular towards languages with little parallel data

Zero-shot translation

- Train on Portuguese↔English and English↔Spanish
- The model can translate Portuguese→Spanish without having seen Portuguese–Spanish sentence pairs



- The shared representation acts as a language-independent meaning space
- Compare the **interlingua** of the Vauquois triangle: here the shared representation is learned from data rather than designed by hand

MT with large language models

- For a large language model, translation is one instruction among others
  - *Translate into Dutch, keeping the informal tone: "Hey, wanna grab lunch?"*
- No dedicated translation training: the ability results from multilingual pretraining and instruction tuning
- Compared to dedicated MT systems:
  - **document-level** context and consistency
  - control over **style and register** through the prompt
  - terminology constraints, explanations, interactive post-editing

Course evaluation: project

- Model design, implementation, and evaluation for an NLP task or a (computational) linguistic research question
- Two models from two different paradigms (symbolic, statistical, neural)
- Evaluation of both models, and a comparative discussion of performance and interpretability
- Two options:
  - **Toxic Comment Classification** (dataset provided)
  - **Choose your own dataset** (emotion, news topics, sarcasm, POS, NER, spam, another language, ...)

Project: requirements

- Fixed train/dev/test split; the test set is used once, at the end
- A simple baseline (majority class, rules) next to the two models
- Both models are trained (or written) by you, on the same data, with the same metrics; choices (features, thresholds, hyperparameters) are made on the development data
- Error analysis with concrete examples from your own test set
- Every claim in the discussion is backed by an output in the notebook
- Generative AI (coding assistants, chatbots) may be used and must be declared; you are responsible for every line of code and every claim

4 ECTS vs. 6 ECTS version

- 4 ECTS: as above
- 6 ECTS: as above, extended with (choice):
  - Deeper discussion of linguistic interpretability
  - Thorough discussion of application for digital humanities research
- Deliverables: a notebook that runs, with saved outputs; optionally a separate report
- Deadline: Tuesday 12 January 2027